

Derivatives of elementary functions



Outline

Constant function and power functions

Derivatives of elementary trigonometric functions

Derivatives of the exponential and logarithmic functions with base e

Constant function and power functions

Introduction: Constant function and power functions

- ▶ In what follows we derive the derivative functions of some of the main **elementary functions**.
- ▶ The tool is always the same: the **definition of derivative** as the limit of the difference quotient

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

- ▶ We start from the constant function and the power functions x^n and x^α : we will use the binomial theorem and the fundamental limits.

Derivative of the constant function

Observation 1

Given the function $f(x) = c$ with $c \in \mathbb{R}$ constant, the derivative function is $f'(x) = 0$.

Proof

We have:

$$\frac{f(x+h) - f(x)}{h} = \frac{c - c}{h} = 0$$

hence

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} 0 = 0$$



Derivative of the function x^n

Observation 2

Given the function $f(x) = x^n$ with $n \in \mathbb{N}, n \geq 1$, the derivative function is $f'(x) = n x^{n-1}$.

Proof

From the binomial theorem (Newton's binomial formula) we have:

$$\begin{aligned}\frac{f(x+h) - f(x)}{h} &= \frac{(x+h)^n - x^n}{h} = \frac{\sum_{k=0}^n \binom{n}{k} x^{n-k} h^k - x^n}{h} \\&= \frac{x^n + n x^{n-1} h + \sum_{k=2}^n \binom{n}{k} x^{n-k} h^k - x^n}{h} \\&= n x^{n-1} + \underbrace{\frac{\sum_{k=2}^n \binom{n}{k} x^{n-k} h^k}{h}}_{\rightarrow 0}\end{aligned}$$

hence

$$f'(x) = \lim_{h \rightarrow 0} n x^{n-1} + \underbrace{\frac{\sum_{k=2}^n \binom{n}{k} x^{n-k} h^k}{h}}_{\rightarrow 0} = n x^{n-1}$$



Proof: the case $n = 2$

With $n = 2$, we have:

$$\frac{f(x+h) - f(x)}{h} = \frac{(x+h)^2 - x^2}{h} = \frac{x^2 + 2xh + h^2 - x^2}{h} = 2x + h$$

hence

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} 2x + h = 2x$$



Derivative of the function x^α

Observation 3

Given the function $f(x) = x^\alpha$ with $\alpha \in \mathbb{R}$, the derivative function is $f'(x) = \alpha x^{\alpha-1}$ for $x > 0$.

Proof (1/2)

Let $x > 0$. We have:

$$\begin{aligned}\frac{f(x+h) - f(x)}{h} &= \frac{(x+h)^\alpha - x^\alpha}{h} = \frac{\left(x \left(1 + \frac{h}{x}\right)\right)^\alpha - x^\alpha}{h} \\ &= x^\alpha \cdot \frac{\left(1 + \frac{h}{x}\right)^\alpha - 1}{h} \\ &\sim x^\alpha \cdot \frac{\alpha \frac{h}{x}}{h} = \alpha x^{\alpha-1} \quad \text{for } h \rightarrow 0\end{aligned}$$

where we used the fundamental limit

$$(1 + \varepsilon(h))^\alpha - 1 \sim \alpha \varepsilon(h) \quad \text{for } \varepsilon(h) \rightarrow 0$$

Proof (2/2)

where

$$\varepsilon(h) = \frac{h}{x} \rightarrow 0 \quad \text{for } h \rightarrow 0.$$

Hence

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \alpha x^{\alpha-1} = \alpha x^{\alpha-1}$$



Example: derivative function

$$f(x) = x^{10} \qquad f'(x) = 10 x^9 \quad (\text{for } x \in \mathbb{R})$$

$$f(x) = \frac{1}{x} = x^{-1} \qquad f'(x) = -\frac{1}{x^2} \quad (\text{for } x > 0)$$

$$f(x) = \sqrt{x} = x^{\frac{1}{2}} \qquad f'(x) = \frac{1}{2\sqrt{x}} \quad (\text{for } x > 0)$$

Constant function and power functions:

- ▶ $f(x) = c \Rightarrow f'(x) = 0$
- ▶ $f(x) = x^n, n \in \mathbb{N}, n \geq 1 \Rightarrow f'(x) = n x^{n-1}$ (binomial theorem)
- ▶ $f(x) = x^\alpha, \alpha \in \mathbb{R} \Rightarrow f'(x) = \alpha x^{\alpha-1}$ for $x > 0$ (fundamental limit of $(1 + \varepsilon)^\alpha - 1$)

Derivatives of elementary trigonometric functions



Introduction: Elementary trigonometric functions

- ▶ In this section we compute the derivatives of $\sin x$ and $\cos x$.
- ▶ We will use the **addition formulas** and the fundamental limits

$$\frac{1 - \cos h}{h^2} \rightarrow \frac{1}{2}, \quad \frac{\sin h}{h} \rightarrow 1 \quad \text{for } h \rightarrow 0$$

- ▶ Trigonometric functions are the basis for describing periodic phenomena: knowing their derivatives is fundamental.

Derivative of the sine function

Observation 4

Given the function $f(x) = \sin x$, the derivative function is $f'(x) = \cos x$.

Proof (1/3)

Using the addition formulas we have:

$$\begin{aligned}\frac{f(x+h) - f(x)}{h} &= \frac{\sin(x+h) - \sin x}{h} \\&= \frac{\sin x \cos h + \sin h \cos x - \sin x}{h} \\&= \sin x \frac{\cos h - 1}{h} + \frac{\sin h}{h} \cos x\end{aligned}$$

Using the fundamental limit

$$\frac{1 - \cos h}{h^2} \rightarrow \frac{1}{2} \quad \text{for } h \rightarrow 0$$

Proof (2/3)

we have

$$\frac{\cos h - 1}{h} = h \cdot \left(\underbrace{-\frac{1 - \cos h}{h^2}}_{\rightarrow -\frac{1}{2} \text{ for } h \rightarrow 0} \right) \sim -\frac{1}{2} h \quad \text{for } h \rightarrow 0.$$

Using the fundamental limit

$$\frac{\sin h}{h} \rightarrow 1 \quad \text{for } h \rightarrow 0$$

we have

$$\frac{\sin h}{h} \cos x \sim \cos x \quad \text{for } h \rightarrow 0.$$

Proof (3/3)

Then

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \sin x \frac{\cos h - 1}{h} + \frac{\sin h}{h} \cos x \\ &= \lim_{h \rightarrow 0} \sin x \left(-\frac{1}{2} h \right) + \cos x = \cos x \end{aligned}$$



Derivative of the cosine function

Observation 5

Given the function $f(x) = \cos x$, the derivative function is $f'(x) = -\sin x$.

Proof (1/2)

Using the addition formulas we have:

$$\begin{aligned}\frac{f(x+h) - f(x)}{h} &= \frac{\cos(x+h) - \cos x}{h} \\ &= \frac{\cos x \cos h - \sin x \sin h - \cos x}{h} \\ &= \cos x \frac{\cos h - 1}{h} - \frac{\sin h}{h} \sin x\end{aligned}$$

Proof (2/2)

Using the arguments of the previous proof, we have:

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \cos x \frac{\cos h - 1}{h} - \frac{\sin h}{h} \sin x \\ &= \lim_{h \rightarrow 0} \cos x \left(-\frac{1}{2} h \right) - \sin x = -\sin x \end{aligned}$$



Summary

Elementary trigonometric functions:

- ▶ $f(x) = \sin x \Rightarrow f'(x) = \cos x$
- ▶ $f(x) = \cos x \Rightarrow f'(x) = -\sin x$
- ▶ Tools: addition formulas and fundamental limits $\frac{1-\cos h}{h^2} \rightarrow \frac{1}{2}$,
 $\frac{\sin h}{h} \rightarrow 1$

Derivatives of the exponential and logarithmic functions with base e



Introduction: Exponential and logarithm with base e

- ▶ In this section we compute the derivatives of e^x and of $\log x$.
- ▶ We will use the fundamental limits

$$\frac{e^h - 1}{h} \rightarrow 1, \quad \log(1 + \varepsilon) \sim \varepsilon \quad \text{for } h, \varepsilon \rightarrow 0$$

- ▶ We will then apply the results to compute **tangent lines** to the graph of a function.

Derivative of the exponential function

Observation 6

Given the function $f(x) = e^x$, the derivative function is $f'(x) = e^x$.

Proof

We have

$$\frac{f(x+h) - f(x)}{h} = \frac{e^{x+h} - e^x}{h} = e^x \cdot \frac{e^h - 1}{h} \sim e^x \quad \text{for } h \rightarrow 0$$

using the fundamental limit

$$\frac{e^h - 1}{h} \rightarrow 1 \quad \text{for } h \rightarrow 0,$$

hence

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} e^x = e^x$$



Derivative of the logarithm function

Observation 7

Given the function $f(x) = \log x$, the derivative function is $f'(x) = \frac{1}{x}$.

Proof (1/2)

We have:

$$\begin{aligned}\frac{f(x+h) - f(x)}{h} &= \frac{\log(x+h) - \log x}{h} = \frac{\log\left(1 + \frac{h}{x}\right)}{h} \\ &\sim \frac{h}{x} \cdot \frac{1}{h} = \frac{1}{x} \quad \text{for } h \rightarrow 0\end{aligned}$$

Proof (2/2)

We used the fundamental limit

$$\log(1 + \varepsilon(h)) \sim \varepsilon(h) \quad \text{for } \varepsilon(h) \rightarrow 0,$$

and

$$\varepsilon(h) = \frac{h}{x} \rightarrow 0 \quad \text{for } h \rightarrow 0,$$

hence

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{1}{x} = \frac{1}{x}$$



Example: tangent line (1/2)

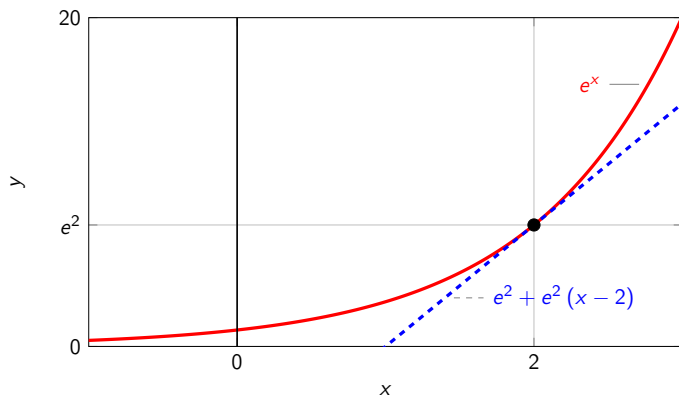
Let us compute the equation of the tangent line to the graph of the function

$$f(x) = e^x \text{ at the point with abscissa } x = 2$$

We have $f(2) = e^2$, $f'(x) = e^x$, $f'(2) = e^2$, hence the tangent line at the point $(2, e^2)$ is:

$$y = f(2) + f'(2)(x - 2) = e^2 + e^2 (x - 2)$$

Example: tangent line (2/2)



Example: tangent line (1/2)

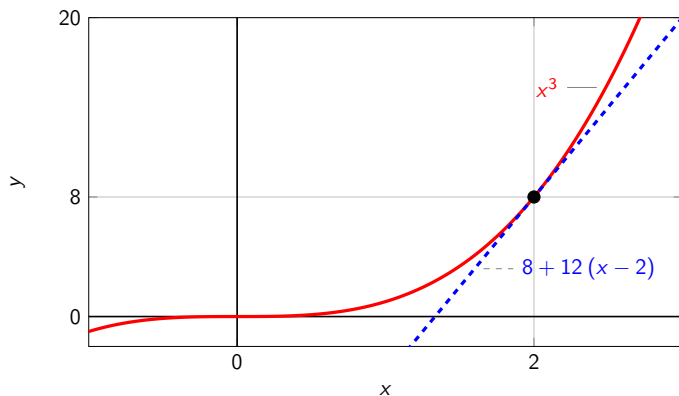
Let us compute the equation of the tangent line to the graph of the function

$$f(x) = x^3 \text{ at the point with abscissa } x = 2$$

We have $f(2) = 8$, $f'(x) = 3x^2$, $f'(2) = 12$, hence the tangent line at the point $(2, 8)$ is:

$$y = f(2) + f'(2)(x - 2) = 8 + 12(x - 2)$$

Example: tangent line (2/2)



Exponential and logarithm with base e :

- ▶ $f(x) = e^x \Rightarrow f'(x) = e^x$
- ▶ $f(x) = \log x \Rightarrow f'(x) = \frac{1}{x}$
- ▶ Tangent line to the graph at the point with abscissa x_0 :
$$y = f(x_0) + f'(x_0)(x - x_0)$$

Key messages

Summary

Derivatives of elementary functions:

$f(x)$	$f'(x)$
c	0
$x^n \ (n \in \mathbb{N}, n \geq 1)$	$n x^{n-1}$
$x^\alpha \ (\alpha \in \mathbb{R})$	$\alpha x^{\alpha-1} \quad \text{for } x > 0$
$\sin x$	$\cos x$
$\cos x$	$-\sin x$
e^x	e^x
$\log x$	$\frac{1}{x}$

- All of them follow from the definition of derivative and from the fundamental limits.